

EDUCATION

University of Maryland, Baltimore County (UMBC) <i>Ph.D. in Information Systems: Collaborative Synthesis of Data Systems</i>	Baltimore, USA Aug. 2022 – May 2026
Cologne University of Applied Sciences (Cologne UAS) <i>M.Sc. in Computer Science / Information Systems; grade 1.2/5 equiv. GPA: 3.8/4</i>	Cologne, Germany Mar. 2019 – Apr. 2022
Macquarie University (MU) <i>Exchange semester in Information Systems; grades HD, HD, C equiv. GPA: 3.2/4</i>	Sydney, Australia June. 2017 – Dec. 2017
Cologne University of Applied Sciences (Cologne UAS) <i>B.Sc. in Business Information Systems; grade 1.7/5 equiv. GPA: 3.3/4</i>	Gummersbach, Germany Aug. 2015 – Feb. 2019

PROFESSIONAL EXPERIENCE

Big Data Analytics Department at Cologne UAS <i>Research Associate (On-site)</i>	Cologne, Germany Oct. 2023 - Present
• Research on Data Management using Machine Learning and Language Models • Summer 2025, Winter 2025/2026, Summer 2026: Teaching Assistant for Databases I and II • Supervision of undergraduate theses and projects	
Dortmund University of Applied Sciences and Arts <i>Lecturer (Hybrid)</i>	Dortmund, Germany Oct. 2023 - Apr. 2025
• Fall 2023 and Fall 2024: Programming in Data Science with Python and R (Programming II Syllabus)	
Information Systems Department at UMBC <i>Graduate Teaching and Research Assistant (Hybrid)</i>	Baltimore, USA Aug. 2022 - Jan. 2026
• Spring 2024: Semantic Mapper for Application Integration (Technology Catalyst Fund) • Fall 2022, Spring 2023, Fall 2024, Spring 2025, Fall 2025: Laboratories for Extended Databases (IS420 and IS620) • Fall 2022: Guest lecture about Inteplato in Systems and Information Integration (IS722)	
IT Consultancy (BWI GmbH) for Bundeswehr (German Armed Forces) <i>Data Warehouse Application Consultant (On-site)</i>	Bonn, Germany Mar. 2019 - Jul. 2022
• Developed data pipelines for materials, missions, and unit-critical dashboards on BW/4HANA for >300K users • Designed role authorization, testing routines, training materials, and workshop leader of >20 participants	

AWARDS AND SERVICES

VLDB Summer School (2023) awarded with **Best Poster** in Cluj, Romania

UMBC Graduate Student Association Conference Travel Grant for CSCI (2022) in Las Vegas, USA

UMBC Graduate Student Representative for Tenure and Promotion Committee (2022) in Baltimore, USA

Oracle Industry Speaker at DOAG (2022) “Machine Learning for Data Integration” in Duesseldorf, Germany

Fulbright Germany Graduate Scholarship at UMBC (2021) in Baltimore, USA

Reaccreditation Auditor for “M.Sc. Digital Sciences” at Cologne UAS (2019 - 2022) in Cologne, Germany

Foerdereverein Gummersbach Study Abroad Funds at MU (2017) in Sydney, Australia

PROGRAMMING SKILLS

SQL, Python (numpy, pandas, matplotlib, sklearn, keras), R, Javascript (npm, vite, react), MATLAB, ABAP

PUBLICATIONS

Khwaileh, Enas; **Traeger, Leonard**; Albert, Gatt; Yannis, Velegrakis. TABERTA: A Language Model for Dataset Discovery. 30th International Conference on Extending Database Technology (EDBT 2027).

[DOI:10.48786/edbt.2027.02](https://doi.org/10.48786/edbt.2027.02) [HuggingFace](#)

Traeger, Leonard; Khwaileh, Enas; Behrend, Andreas; Karabatis, George. RACT: Retrieval Augmented Column-Table Learning for Multi-Table Schema Matching. Technical Report (ArXiv 2026).

<https://arxiv.org/abs/2606.07843> [GitHub](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. Scoper: Streamline Linkable Schemas for Matching. IEEE 42st International Conference on Data Engineering (ICDE 2026). (to appear in proceedings) [GitHub](#)

[YouTube Demo Application](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. Collaborative Scoping: Self-Supervised Linkability Assessment for Schema Matching. 29th International Conference on Extending Database Technology (EDBT 2026). [DOI:10.48786/edbt.2026.03](https://doi.org/10.48786/edbt.2026.03) [GitHub](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. Collective Scoping: Streamlining Entity Sets Towards Efficient and Effective Entity Linkages. Springer Nature Computer Science Volume 6 (SNCS 2025).

[DOI:10.1007/s42979-025-03734-7](https://doi.org/10.1007/s42979-025-03734-7) [GitHub](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. SEALM: Semantically Enriched Attributes with Language Models for Linkage Recommendation. 27th International Conference on Enterprise Information Systems (ICEIS 2025). [DOI:10.5220/0013217700003929](https://doi.org/10.5220/0013217700003929) [GitHub](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. Scoping: Towards Streamlined Entity Collections for Multi-Sourced Entity Resolution with Self-Supervised Agents”. 26th International Conference on Enterprise Information Systems (ICEIS 2024). **Best Paper Award** [DOI:10.5220/0012607500003690](https://doi.org/10.5220/0012607500003690) [GitHub](#)

Traeger, Leonard; Omar, Faraque. Autonomous Object Detection and Landing with Parrot Rolling Spider in Matlab Simulink”. IEEE International Conference on Embedded Systems, Cyber-physical Systems & Applications (ESCS 2023). [DOI 10.1109/CSCE60160.2023.00160](https://doi.org/10.1109/CSCE60160.2023.00160) [GitHub](#) [YouTube](#)

Traeger, Leonard; Behrend, Andreas; Karabatis, George. Inteplato: Generating Mappings of Heterogeneous Relational Schemas using Unsupervised Learning”. IEEE International Conference on Computational Science and Computational Intelligence (CSCI 2022). [DOI:10.1109/CSCI58124.2022.00083](https://doi.org/10.1109/CSCI58124.2022.00083) [GitHub](#)

PATENTS

Karabatis, George; Behrend, Andreas; **Traeger, Leonard**. Systems and methods for organizing data of different local data schemas based on similarity rankings, scoring, and signatures. [US Patent App. 18/600,828](#) (2024). (granted)